

Vulnerability Triage Challenge

Module 5 In-Class Activity · Tuesday, March 24, 2026 · Not Graded

Name(s): Milos Petrovic **Partner:** None

Scenario — Ridgeline Regional Medical Center

You are a security analyst at Ridgeline Regional Medical Center (200-bed community hospital). HIPAA requires protection of all PHI. A 2-hour maintenance window opens TONIGHT. Your supervisor needs to know: which 3 findings must be fixed now, and how to schedule the rest.

#	CVE / Plugin	Vulnerability	System	CVSS	Severity	Internet-Facing?
A	CVE-2021-44228	Log4Shell – Apache Log4j RCE	EHR App Server (internal)	10.0	Critical	No
B	CVE-2018-15919	OpenSSH Username Enumeration	Linux File Server (internal)	5.3	Medium	No
C	CVE-2021-1675	PrintNightmare – Print Spooler RCE	Nurse Workstations (×12)	7.8	High	No
D	Plugin 51192	SSL Self-Signed Certificate	Patient Portal (internet)	6.5	Medium	Yes
E	CVE-2022-22965	Spring4Shell – Spring Framework RCE	Patient Portal (internet)	9.8	Critical	Yes
F	Plugin 57608	SMB Signing Disabled	All Windows hosts (×15)	5.9	Medium	No
G	CVE-2019-11510	Pulse Secure VPN – Arbitrary File Read	VPN Gateway (internet)	10.0	Critical	Yes
H	Plugin 10863	TLS 1.0/1.1 Enabled	Patient Portal (internet)	3.7	Low	Yes

Fill in 'Internet-Facing?' column first: Yes / No / Partial — this affects your analysis.

Task 1 — Your Top 3 Immediate Fixes (Tonight's Window)

Identify the 3 findings you must address tonight. For each, write 2–3 sentences justifying your choice (CVSS score, exposure, data at risk).

Fix #1 — Finding: _____

Justification: Log4Shell has a CVSS of 10.0 and affects the internal HER server, which stores all patients PHI. Active exploits have been publicly available since December 2021, making this the highest-priority patch. A successful attack would constitute a HIPAA breach and could disrupt patient care system entirely.

Fix #2 — Finding: _____

Justification:

Spring4Shell scores 9.8 and allows remote code execution on the internet-facing patient wab portal. An external attacker can exploit this without authentication, gaining direct access to patient data. Patching the Spring Framework or applying a WAF virtual patch is feasible within the 2-hour window.

Fix #3 — Finding: _____

Justification:

The Pulse Secure VPN vulnerability scores 10.0 and is internet-facing, making it an active ransomware entry point. Compromising the VPN gateway gives attackers full access to the internal network. This is single appliance – applying the vendor patch and rebooting fits within the maintenance window.

Task 2 — Ranking the Remaining 5

Priority	Finding	Vulnerability	Timeline	Scheduling Rationale
#4	C	PrintNightmare	Urgent	RCE on 12 nurse workstations with access to medication systems, compensating control applied tonight, patch this week

#5	F	SMB Signing Disabled	Urgent	Affects all 15 Windows hosts, enables NTLM relay and lateral movement across the entire network.
#6	D	SSL Self-Signed Certificate	Planned	Internet-facing portal, enables MITM attacks and erodes patient trust, low effort to replace with valid cert
#7	B	OpenSSH Username Enumeration	Planned	Internal only, no direct RCE but enables targeted brute-force attacks against the file server
#8	H	TLS 1.0/1.1 Enabled	Scheduled	Deprecated protocols with low exploitability, attacker needs privileged network position to abuse

Timeline key: Urgent = this week · Planned = within 30 days · Scheduled = 90 days

Task 3 — Compensating Control for Finding C (PrintNightmare)

You cannot patch 12 nurse workstations tonight. What compensating control reduces risk until you can schedule the patches? Be specific — name the control, how to implement it, and the risk it addresses.

Disable the Windows Print Spooler service on all 12 nurse workstations tonight using Group Policy or remote PowerShell. Since PrintNightmare exploits the Spooler service directly, disabling it eliminates the attack surface without requiring a reboot patch. This is Microsoft's own recommended mitigation. Nurses temporarily lose local printing but can still use server-based print queues. The patch should be applied within the week.

Debrief Notes

Use this space to capture key discussion points from the class debrief:
